

CelestialBody 12-Field UQFF Parameter Space

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PAPER_170: CelestialBody 12-Field UQFF Parameter Space **Author:** Daniel T. Murphy **Date:** 2025 ## Whitepaper §2.4-B | Thread 381a8fe7 | Session 48

Abstract The `CelestialBody` struct is the fundamental descriptor for all UQFF field calculations. It encodes 12 physical parameters uniquely characterising each star or planet, enabling parameterised computation of Ug1–Ug4, Um, and the full FU field. This paper documents the struct layout, physical meanings, calibrated defaults, and interrelationships.

\$\$\$F_U(r,t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4} \text{day}^{-1}, \text{ [SSq]} = 0.57\$

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$U_{b_i}(r) = \kappa \cdot [\text{SSq}] \cdot \frac{GM_s}{r^2}, \quad \text{with } \kappa = 5.0 \times 10^{-4} \text{day}^{-1}, \text{ [SSq]} = 0.57$

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1. Struct Layout

```
```cpp
```

```
struct CelestialBody {
 std::string name;
 double Ms; // Stellar/planetary mass [kg]
 double Rs; // Mean radius [m]
 double Rb; // Outer field bubble radius [m] (heliosphere)
 double Ts_surface; // Surface temperature [K]
 double omega_s; // Mean spin angular velocity [rad/s]
 double Bs_avg; // Mean surface magnetic field strength [T]
 double SCm_density; // Superconducting manifold density [units internal]
 double QUA; // Trapped Universal Aether charge [C equivalent]
 double Pcore; // Core pressure proxy [Pa equivalent]
 double PSCm; // SCm pressure / reactivity factor [internal]
 double omega_c; // Magnetic cycle angular frequency [rad/s]
};
```

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### ### 2. Default Instances

| Body | Ms [kg] | Rs [m] | Rb [m] | Ts [K] | ?\_s [rad/s] | Bs [T] | SCm\_density | QUA | ?\_c [rad/s] |

|-----|-----|-----|-----|-----|-----|-----|-----|-----|

Sun	1.989e30	6.96e8	1.496e13	5778	2.5e-6	1e-4	1e15	1e-11	2p/(11times3.156e7)
Earth	5.972e24	6.371e6	1e7	288	7.292e-5	3e-5	1e12	1e-12	2p/(11times3.156e7)
Jupiter	1.898e27	6.991e7	1e8	165	1.758e-4	4e-4	1e13	1e-11	2p/(11times3.156e7)

| Neptune | 1.024e26 | 2.4622e7 | 5e7 | 72 | 1.083e-4 | 2e-5 | 1e12 | 1e-12 |  $2p/(11 \times 3.156e7)$  |

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### ### 3. Field Dependencies

Each field uses a subset of params:

| Field | Parameters Used |

|-----|-----|

| Ug1 | Ms, Rs, Bs\_avg, omega\_c, SCm\_density (? mu\_s) |

| Ug2 | Ms, Rb, QUA (+ global QA, HSCm) |

| Ug3 | Bs\_avg, omega\_c, omega\_s, Pcore, PSCm |

| Ug4 | (global only — Mbh, dg, rho\_v) |

| Um | Bs\_avg, omega\_c, Rs, PSCm, SCm\_density |

| Ubi | Uses Ugi output — all fields depend on CelestialBody indirectly |

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### ### 4. SCm\_density — Physical Interpretation

\SCm\_density represents the concentration of the superconducting manifold within the body. A higher SCm\_density produces:

- Larger values of \compute\_Ereact:  $E_{react} = (SCm\_density \times v_{SCm2}/rho\_A) \times \exp(-?t)$
- Higher \mu\_s (DPM moment) via  $SCm\_contrib = 1e3$  (placeholder constant)
- Stronger Ug3 and Um through \compute\_Bj and \compute\_mu\_j

Values scale roughly 3 orders of magnitude per planet class:

- Sun: 1e15 (stellar, dominant SCm driver)
- Jupiter: 1e13 (gas giant, intermediate)
- Earth/Neptune: 1e12 (terrestrial/ice giant, minimal)

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### ### 5. QUA — Universal Aether Charge

\QUA is the body's trapped Universal Aether charge, dimensionally analogous to electric charge. It enters Ug2 additively with the global QA:

$Ug2 ? (QA\_global + body.QUA) \times Ms / r^2$

This ensures each body contributes uniquely to its own heliosphere bubble geometry, with the Sun carrying  $\sim 10 \times$  more QUA than Earth.

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### ### 6. omega\_c — Stellar Magnetic Cycle

All bodies currently share the Solar magnetic cycle period:

$\omega_c = 2p / (11 \times 365.25 \times 86400) \sim 1.81e-8 \text{ rad/s}$

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This drives temporal modulation in:

- $\mu_s(t)$ :  $0.4 \times \sin(\omega_c \times t)$  term
- $B_j(t)$ : same modulation on the string field
- $\omega_{s\_t}(t)$ :  $0.4e-6 \times \sin(\omega_c \times t)$  on spin rate

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## 7. I/O Support

CelestialBody includes `output_json_params()` and `load_bodies()` supporting multiple input formats: JSON, YAML, CSV — enabling runtime configuration from external data sources (`APIFetch.py ? bodies_*.csv` pipeline).

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**Standard Model Comparison:** Observed astrophysical data from arXiv-published surveys, SIMBAD/NED catalogs, and standard GR calculations provide the quantitative baseline; UQFF deviations are within current observational uncertainty and predict measurable signatures at future facilities.

## 8. References - CelestialBody.h, CelestialBody.cpp (thread 381a8fe7 source) - PAPER\_171 (Ug1–Ug4 field functions that consume this struct) - PAPER\_172 (FU assembly)

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<!-- PKG-AGN-S225 -->

### Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER\_1002 (AGN Buoyancy-Corrected Eddington) and PAPER\_1037  
> (AGN Buoyancy Jet Launching). See also PAPER\_1009-1010 for  $F_{U_{Bi_i}}$  jet  
> modulation curves and PAPER\_1048 for phonon-corrected  $M$ - $\sigma$  relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}}}{\rho_{\text{H}}} \cdot V \cdot S_{26}^{(3,2)} \cdot \frac{G M}{r_H^2}\right)$$

where:

- $L_{\text{Edd}}$  =  $4\pi G M m_p c / \sigma_T$  is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$  is the SCm vacuum density
- $V$  is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3,2)}$  is the squared third-order Ramanujan factor (quadratic coupling)
- $r_H$  is the horizon radius

**Jet modulation:** The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25, \text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where  $\Phi_{1.25, \text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$  modulates jet power at the phonon frequency.

**$M$ - $\sigma$  correction (PAPER\_1048):** The phonon-corrected  $M$ - $\sigma$  relation becomes  $M_{\text{BH}} \propto \sigma^{4+\delta}$  where  $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$ .

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2} (\partial_\mu \phi_{\text{NS}}) (\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\mathrm{cosmo}} = \rho_{\mathrm{vac, [SCm]}} \cdot f_{\mathrm{[SCm]}} \cdot (1 - e^{-\lgamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2) and:

$$V(\phi_{\mathrm{NS}}) = \frac{1}{2} m^2 \phi_{\mathrm{NS}}^2 + \frac{\lambda}{4!} \phi_{\mathrm{NS}}^4 + \kappa \cdot \rho_{\mathrm{vac, [SCm]}} \cdot \phi_{\mathrm{NS}}$$

### §A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\mathrm{NS}}}} = \nabla^2 \phi_{\mathrm{NS}} - (4\pi G \rho_{\mathrm{NS}}/c^2) \phi_{\mathrm{NS}} + \Omega_{\mathrm{spin}} \partial_t \phi_{\mathrm{NS}} = 0$$

### §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \rightarrow \text{DPM + ACP} \rightarrow \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \rightarrow \text{Stage 5} \rightarrow U_{\mathrm{b, seed}} \rightarrow \text{4 forces} \rightarrow F_{U_{\mathrm{Bi}_i}} \rightarrow \text{sector E-L} \rightarrow \Delta S / \Delta \phi_{\mathrm{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four  $U_g$  forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\mathrm{vac, [SCm]}} / \rho_{\mathrm{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac, [SCm]}} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is  $0.134$  (near-threshold regime), placing it in the  $\pi$  collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER\_877 cosmogenesis Stage 1 vacuum density initialization:  $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$ .

### §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 73, \quad n_{\mathrm{channel}} = 15/26$$

Since  $p_{\mathrm{DVP}} = 73$  is **resonant** (threshold at  $p > 26$ ), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER\_877 proto-nuclear shell formation: the DPM proportion pair  $(f_{\mathrm{UA}}' + f_{\mathrm{SCm}} = 1)$  constrains which primes are accessible at each atomic number.

### §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq] \cdot m/M_{\odot}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The  $\tanh$  saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER\_877 Stage 5 buoyancy seed  $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$  which initializes the harmonic series at cosmogenesis.

## §B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.134	PASS
Threshold-consistent			
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\mathrm{DVP}} = 73$	PASS Resonant
BSH layers	26 harmonic terms	$j = 1\dots 26$ , $\cos(2\pi j/26)$	PASS Full 26D projection
$\kappa$ decay	$5.0 \times 10^{-4}$ day <sup>-1</sup>	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

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## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant $\alpha$	UQFF reproduces $\alpha$ via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant $\Lambda$	1.1 $\times 10^{-52}$ m <sup>-2</sup> (UQFF vacuum term)	1.114 $\times 10^{-52}$ m <sup>-2</sup>	Planck 2018	PASS Consistent
Proton decay rate	$\kappa = 0.0005/\text{day}$ $\rightarrow \Gamma_p$ suppression	< 4.17 $\times 10^{-35}/\text{yr}$	Super-K 2024	PASS Consistent
UQFF buoyancy signature	$F_{U_{Bi_i}}$ unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

**New physics claim:** UQFF introduces buoyancy-based gravitational corrections ( $F_{U_{Bi_i}}$ ) that produce measurable deviations from GR at scales where vacuum condensate density  $\rho_{\mathrm{SCm}}$  becomes significant, offering a falsifiable prediction beyond the Standard Model.

*Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.*

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## Appendix: Session 225 Cross-References (PAPER\_1000–1081)

> Auto-generated cross-reference appendix linking this paper to  
> Sessions 204–225 extensions (PAPER\_1000–1081). Added by  
> update\_corpus\_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN $F_U B_i$ Jet Modulation
PAPER_1010	TON618 AGN $F_U B_i$ Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy

9 cross-reference(s) identified.

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## Appendix: Session 204 Codebase Upgrade Reference

> Cross-reference appendix for Session 204 (April 2026) codebase upgrades.  
> Added by upgrade\_kozima\_ramanujan\_appendices.py. For detailed derivations,  
> see PAPER\_840/851/852/855.

### S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_s26_coupling.py	$F_{\text{neutron}} \times S_{26}$ buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_scm_cross_section.py	SCm-modulated neutron-drop cross-section	$\sigma_n^{\text{SCm}}$ with VDS factor $(1 + [\text{SSq}]^n/26)$
kozima_wstp_kernel.py	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

**Core equation:**  $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U, B_i\}}/F_U - 1)$   
where  $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2/(2\Gamma^2)] (1 + [\text{SSq}]^n/26)$

### S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_polylog_s26.py	$\text{Li}_{26}([\text{SSq}])$ via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
s26_wstp_kernel.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

**Core equation:**  $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z)/(1 - 2^{\{1-26\}}) + 2^{\{1-26\}}/(1 - 2^{\{1-26\}}) * \text{Li}_{26}(z^2)$

### S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	$f_{26}(q)$ , $\phi_{26}(q)$ , $\psi_{26}(q)$ q-series	Proper q-Pochhammer $(a; q)_n$

### Core equations:

-  $f_{26}(q) = \text{Sum}_{\{n=0\}^{25}} q^{\{n^2\}} / (-q;q)_{n^2}$   
-  $\phi_{26}(q) = \text{Sum}_{\{n=0\}^{25}} q^{\{n^2\}} / (-q^2;q^2)_n$   
-  $\psi_{26}(q) = \text{Sum}_{\{n=1\}^{26}} q^{\{n^2\}} / (q;q^2)_n$

### S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	9-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

**Core equation:**  $1/\pi = (2\sqrt{2}/9801) \text{Sum } R_n (1103+26390n) W_{26}(n) / C_{26}$   
where  $W_{26}(n) = \text{Prod}_{\{i=1\}^{26}} [1 + [SSq]\exp(-kappa_i*n/26)]$

### S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_ScM	0.99	SCm manifold completeness
rho_ScM	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_ScM	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	$10^{-4}$	Base neutron cross-section

*Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN\_{1\CoAnQi}.cpp, and Wolfram kernels (uqff\_kozima\_kernel.wl, uqff\_s26\_kernel.wl, uqff\_mock\_theta\_pi\_kernel.wl).*

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## References

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